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পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED

05 STORIED RESIDENTIAL BUILDING

AT POST- SYLHET SADAR, KOTWALI, SYLHET
DAG # SA-2321, BS-14557, KHA # 4526, MOUJA-BAGBARI

CLIENT

MD. NAZRUL ISLAM

Verification Code: YXqNR5p9
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

THE PERFECT SOIL INVESTIGATOR

Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-65ft	13
02	Disturbed sample	02	0-70ft	14
03	Disturbed sample	03	0-65ft	13

Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α – Adhesion factor, f_{su} – Ultimate skin friction, C_u – Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = 4N/200\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)**

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	3	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	7	0.864	0.909	0.991	1.020	1.102	1.199	0.931
BH02	2.5-5.0	4	0.464	0.488	0.532	0.548	0.592	0.644	0.500
	5.0-10	7	0.816	0.858	0.936	0.964	1.041	1.133	0.879
BH03	2.5-5.0	3	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	9	1.020	1.073	1.169	1.205	1.301	1.416	1.099
BH04	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH05	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

** Designer should consider both the calculated design N value and settlement behavior of the corresponding soil profile while selecting shallow footings

Table 8: The allowable skin friction for precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.074	0.049	0.025	-	-	-	-	-	-	-
10	0.096	0.138	0.118	-	-	-	-	-	-	-
15	0.049	0.074	0.049	-	-	-	-	-	-	-
20	0.128	0.261	0.152	-	-	-	-	-	-	-
25	0.184	0.168	0.248	-	-	-	-	-	-	-
30	0.096	0.138	0.138	-	-	-	-	-	-	-
35	0.074	0.049	0.049	-	-	-	-	-	-	-
40	0.096	0.049	0.074	-	-	-	-	-	-	-
45	0.184	0.049	0.074	-	-	-	-	-	-	-
50	0.096	0.096	0.096	-	-	-	-	-	-	-
55	0.192	0.156	0.468	-	-	-	-	-	-	-
60	0.136	0.321	0.669	-	-	-	-	-	-	-
65	0.669	0.669	0.669	-	-	-	-	-	-	-
70	-	0.669	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for precast driven piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	0.450	0.225	-	-	-	-	-	-	-
10	0.900	1.350	1.125	-	-	-	-	-	-	-
15	0.450	0.675	0.450	-	-	-	-	-	-	-
20	25.600	2.700	30.400	-	-	-	-	-	-	-
25	36.800	33.600	49.600	-	-	-	-	-	-	-
30	0.900	1.350	1.350	-	-	-	-	-	-	-
35	0.675	0.450	0.450	-	-	-	-	-	-	-
40	0.900	0.450	0.675	-	-	-	-	-	-	-
45	36.800	0.450	0.675	-	-	-	-	-	-	-
50	0.900	0.900	0.900	-	-	-	-	-	-	-
55	38.400	1.575	7.875	-	-	-	-	-	-	-
60	27.200	5.400	11.250	-	-	-	-	-	-	-
65	11.250	11.250	11.250	-	-	-	-	-	-	-
70	-	11.250	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.032	0.016	-	-	-	-	-	-	-
10	0.062	0.090	0.077	-	-	-	-	-	-	-
15	0.032	0.048	0.032	-	-	-	-	-	-	-
20	0.083	0.170	0.099	-	-	-	-	-	-	-
25	0.120	0.109	0.161	-	-	-	-	-	-	-
30	0.062	0.090	0.090	-	-	-	-	-	-	-
35	0.048	0.032	0.032	-	-	-	-	-	-	-
40	0.062	0.032	0.048	-	-	-	-	-	-	-
45	0.120	0.032	0.048	-	-	-	-	-	-	-
50	0.062	0.062	0.062	-	-	-	-	-	-	-
55	0.125	0.101	0.304	-	-	-	-	-	-	-
60	0.088	0.209	0.435	-	-	-	-	-	-	-
65	0.435	0.435	0.435	-	-	-	-	-	-	-
70	-	0.435	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	0.203	0.101	-	-	-	-	-	-	-
10	0.405	0.608	0.506	-	-	-	-	-	-	-
15	0.203	0.304	0.203	-	-	-	-	-	-	-
20	11.520	1.215	13.680	-	-	-	-	-	-	-
25	16.560	15.120	22.320	-	-	-	-	-	-	-
30	0.405	0.608	0.608	-	-	-	-	-	-	-
35	0.304	0.203	0.203	-	-	-	-	-	-	-
40	0.405	0.203	0.304	-	-	-	-	-	-	-
45	16.560	0.203	0.304	-	-	-	-	-	-	-
50	0.405	0.405	0.405	-	-	-	-	-	-	-
55	17.280	0.709	3.544	-	-	-	-	-	-	-
60	12.240	2.430	5.063	-	-	-	-	-	-	-
65	5.063	5.063	5.063	-	-	-	-	-	-	-
70	-	5.063	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five storied residential building.

Soil profile scenario

Overall soil information of the investigated site is very much irregular in between the boreholes. Moreover, vertical inconsistency exists in the present soil formation. In the bore hole 1, clay soil of very soft consistency has been found in the top 15 ft depth followed by alternate layer of clay and sand deposition with a black organic deposition in between 25-35 ft depth. In the bore hole 2, clay soil of very soft to soft consistency has been found up to 55 ft depth followed by stiff clay layer extended up to 70 ft depth with a black organic deposition in between 45-55 ft depth and a dense sand deposition at 25 ft depth. In the bore hole 3, clay soil of very soft to soft consistency has been found up to 50 ft depth followed by stiff clay layer extended up to 65 ft depth with a black organic deposition in between 25-40 ft depth and a dense sand deposition in between 15-25 ft depth. However, soil condition cannot be predicted beyond 65-70 ft depth since no sample is available from that depth.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since the overall soil condition is very erratic in nature and soft cohesive soil with black organic is dominated in the profile piles or piled raft foundation can be the suitable option for the proposed structure. However, it is strongly recommended to perform SPT for another few locations in the proposed site with a depth of up to 100 ft to predict the sol condition more confidently. For designing, the necessary data can be found from the attached table 7, 8, 9, 10 and 11.

Safety provisions

Since the proposed site is entirely dominated with soft cohesive soil with black organic in all the three bore holes, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously, **pile load test** (if provided) covering the entire project area must be carried out for confirming the design parameters of the service pile. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. No other test has been performed by the client.

Special notes

These suggestions are absolutely tentative, and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **65-70 ft** as client provided.

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